EXPERIMENT NO. : 8 (MAIN)

DATE: 08/10/2026

1. AIM

To design a CMOS inverter with 4 fan-out and a CMOS ring oscillator using minimum channel length and width in 180nm technology and:

1] To calculate the propagation delay of CMOS inverter with FO4 and compare it with theoretically calculated delay using logical effort, and compare it with inverter having no load (FO0), FO1, FO2, FO3, and FO5.

2] To find the number of stages N when frequency is 2.22GHz, design and implement it, and compare the observed value of frequency with the desired frequency.

2. TOOL

Cadence Virtuoso

3. THEORY

1. Method of Logical Effort: Logical effort provides a normalized framework to estimate the delay of CMOS logic gates. The delay of a logic gate is modeled as the sum of effort delay (stage effort) and parasitic delay: $d = g \cdot h + p$. Here, g is the logical effort representing the gate's topological complexity, $h = C_{out}/C_{in}$ is the electrical effort (fan-out), and p is the intrinsic parasitic delay. For a reference CMOS inverter, $g = 1$ and $p = 1$. The normalized delay d is converted to absolute time by scaling with the base inverter delay t_{inv}, such that $t_{p,\text{theor}} = d \cdot t_{inv}$.

2. Delay Dependence on Fan-Out: When an inverter drives identical load inverters, each additional load increases the electrical effort h. Since the logical effort and parasitic delay remain fixed, the gate delay increases linearly with the fan-out load. Benchmarking delays from an unloaded state ($h = 0$, FO0) through standard loads like FO4 allows characterization of the drive capability and intrinsic parasitic overhead.

3. CMOS Ring Oscillator Operation: A ring oscillator comprises an odd number N of inverting stages cascaded in a closed loop. Because each stage drives a single identical succeeding stage, the effective load corresponds to a fan-out of 1 (FO1). The closed-loop oscillation period is given by $T = 2 \cdot N \cdot t_p$, where t_p is the per-stage propagation delay. The required stage count for a target oscillation frequency f is therefore governed by $N = \frac{1}{2 \cdot f \cdot t_p}$.

4. TABLES

Table 4.1: Design Parameters

given parameter	value	unit
technology	180	nm
V_{dd}	1.8	V
channel length (L)	180	nm
minimum NMOS width (W_n)	400	nm
symmetric PMOS width (W_p)	800	nm
desired oscillator frequency (f_{des})	2.22	GHz

Table 4.2: Inverter Delay vs. Fan-Out Characterization

fan-out (load)	t_{phl}	t_{plh}	t_p	unit
FO0 (unloaded)	12.6322	16.6138	14.623	ps
FO1	18.9750	27.3618	23.1684	ps
FO2	25.4262	38.7649	32.0956	ps
FO3	31.9120	50.6160	41.264	ps
FO4	38.4404	62.7786	50.6095	ps
FO5	46.1987	71.0818	58.6403	ps

Table 4.3: 7-Stage Ring Oscillator Performance

parameter	value	unit
desired frequency (f_{des})	2.22	GHz
calculated number of stages (N)	7	stages
observed time period (T)	445.9386	ps
observed frequency (f_{obs})	2.2425	GHz
frequency signed error	+1.0135%	%

Table 4.4: Comparison of Theoretical and Practical Values

parameter	theoretical value	practical value	unit	signed error
FO0 propagation delay (t_p)	14.6230	14.623	ps	0%
FO1 propagation delay (t_p)	29.2460	23.1684	ps	-20.781%
FO2 propagation delay (t_p)	43.8690	32.0956	ps	-26.8376%
FO3 propagation delay (t_p)	58.4920	41.264	ps	-29.4536%
FO4 propagation delay (t_p)	73.1150	50.6095	ps	-30.781%
FO5 propagation delay (t_p)	87.7380	58.6403	ps	-33.1643%
oscillator frequency (f)	2.2200	2.2425	GHz	+1.0135%

5. CALCULATION

$$ratio = W_p/W_n = 800/400 = 2$$

$$d_FO0 = g * 0 + p = 1 * 0 + 1 = 1$$

$$T_{p(FO0)(theoratical)} = d_{FO0} * t_{inv} = 1 * 14.623 = 14.623ps$$

$$d_FO1 = g * 1 + p = 1 * 1 + 1 = 2$$

$$T_{p(FO1)(theoratical)} = d_{FO1} * t_{inv} = 2 * 14.623 = 29.246ps$$

$$d_FO2 = g * 2 + p = 1 * 2 + 1 = 3$$

$$T_{p(FO2)(theoratical)} = d_{FO2} * t_{inv} = 3 * 14.623 = 43.869ps$$

$$d_FO3 = g * 3 + p = 1 * 3 + 1 = 4$$

$$T_{p(FO3)(theoratical)} = d_{FO3} * t_{inv} = 4 * 14.623 = 58.492ps$$

$$d_FO4 = g * 4 + p = 1 * 4 + 1 = 5$$

$$T_{p(FO4)(theoratical)} = d_{FO4} * t_{inv} = 5 * 14.623 = 73.115ps$$

$$d_FO5 = g * 5 + p = 1 * 5 + 1 = 6$$

$$T_{p(FO5)(theoratical)} = d_{FO5} * t_{inv} = 6 * 14.623 = 87.738ps$$

$$N = 1000/(2 * f_{des} * t_{FO1})$$

$= 1000/(2 * 2.22 * 29.246)$

$= 7.7010608 \approx 7.7$

$N_{stage} = 7$

$f_{prac} = 1000/T_{obs}$

$= 1000/445.9386$

$= 2.2424612GHz \approx 2.2425GHz$

$Error = (f_{prac} - f_{des})/f_{des} * 100$

$= (2.2424612 - 2.22)/2.22 * 100$

$= 1.0117658\% \approx 1.01\%$

6. CIRCUIT DIAGRAMS & WAVEFORMS

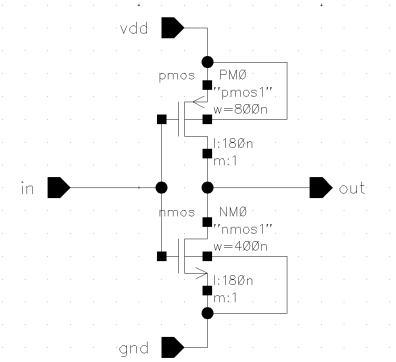


Figure 6.1: Symmetric inverter schematic (
$W_p = 800\text{nm}, W_n = 400\text{nm}$**)**

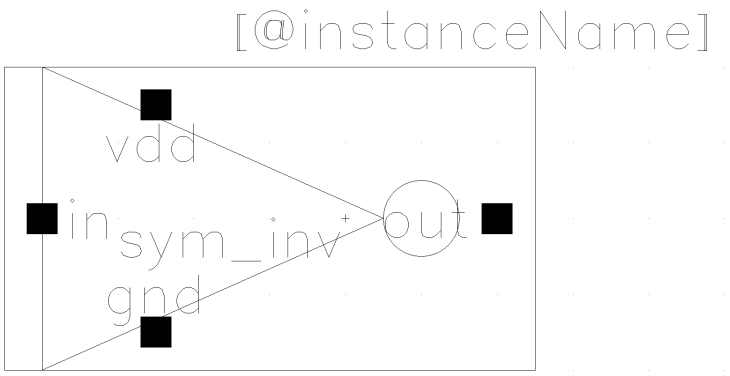


Figure 6.2: Symmetric inverter symbol

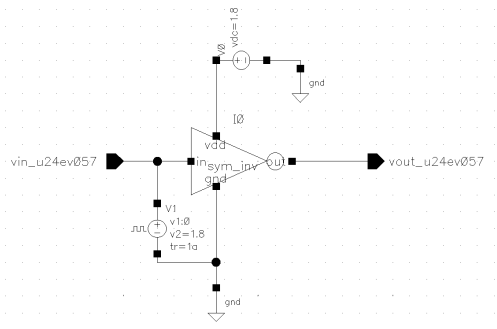


Figure 6.3: Unloaded symmetric CMOS inverter testbench

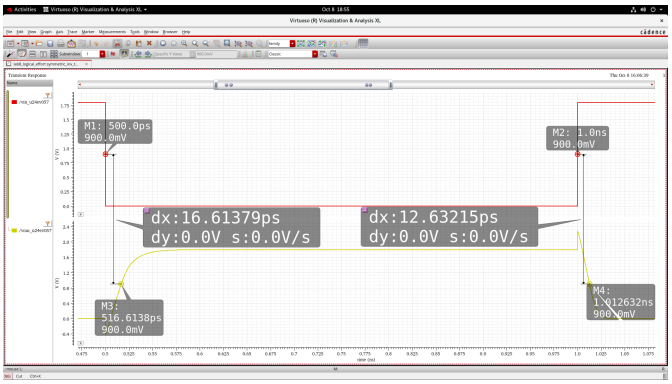


Figure 6.4: Unloaded symmetric inverter waveform (FO0)

$t_{PHL} = 12.6322\text{ps}, \quad t_{PLH} = 16.6138\text{ps}, \quad t_p = 14.6230\text{ps}$

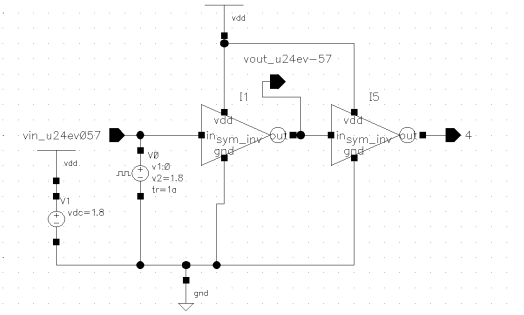


Figure 6.5: CMOS inverter with Fan-Out of 1 (FO1) schematic

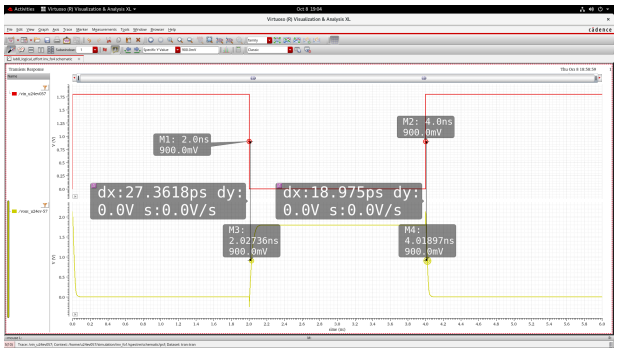


Figure 6.6: CMOS inverter with FO1 transient waveform

$t_{PHL} = 18.9750\text{ps}, \quad t_{PLH} = 27.3618\text{ps}, \quad t_p = 23.1684\text{ps}$

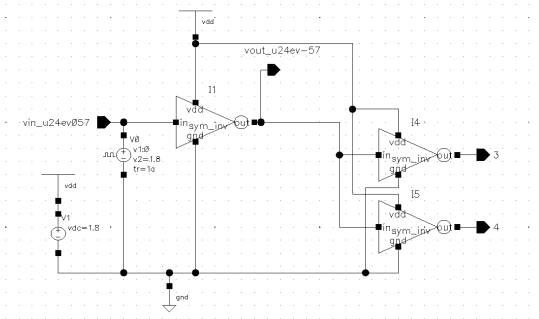


Figure 6.7: CMOS inverter with Fan-Out of 2 (FO2) schematic

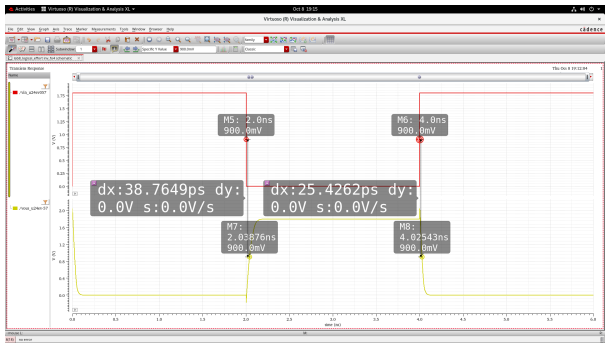


Figure 6.8: CMOS inverter with FO2 transient waveform

$t_{PHL} = 25.4262\text{ps}, \quad t_{PLH} = 38.7649\text{ps}, \quad t_p = 32.0956\text{ps}$

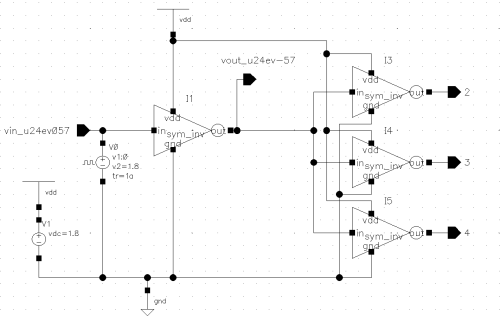


Figure 6.9: CMOS inverter with Fan-Out of 3 (FO3) schematic

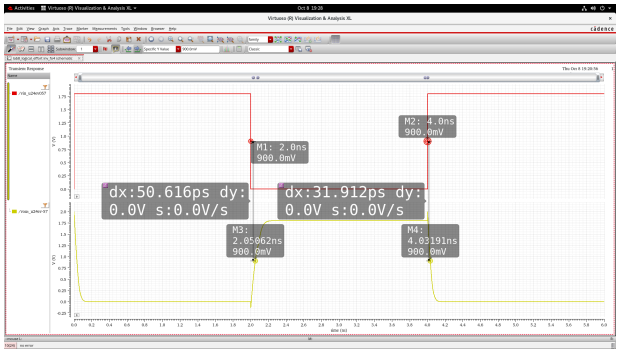


Figure 6.10: CMOS inverter with FO3 transient waveform

$t_{PHL} = 31.9120\text{ps}, \quad t_{PLH} = 50.6160\text{ps}, \quad t_p = 41.2640\text{ps}$

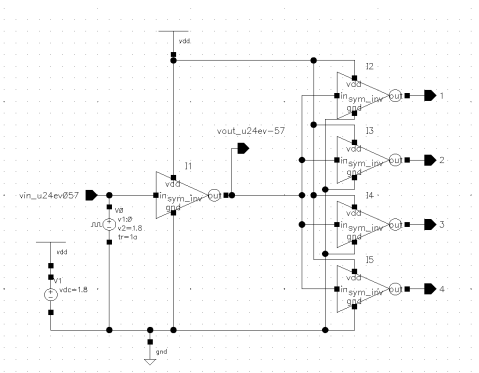


Figure 6.11: CMOS inverter with Fan-Out of 4 (FO4) schematic

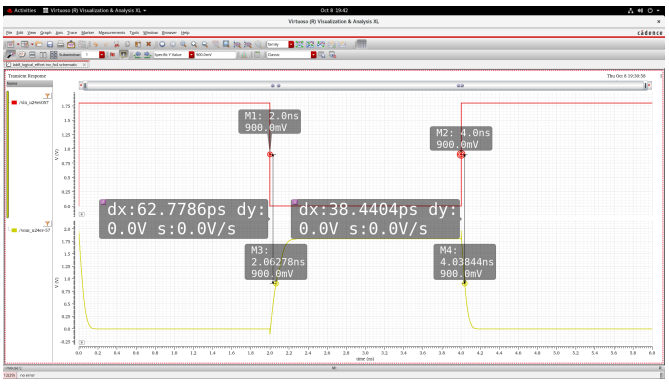


Figure 6.12: CMOS inverter with FO4 transient waveform

$t_{PHL} = 38.4404\text{ps}, \quad t_{PLH} = 62.7786\text{ps}, \quad t_p = 50.6095\text{ps}$

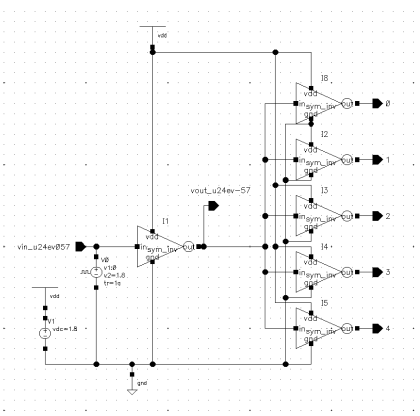


Figure 6.13: CMOS inverter with Fan-Out of 5 (FO5) schematic

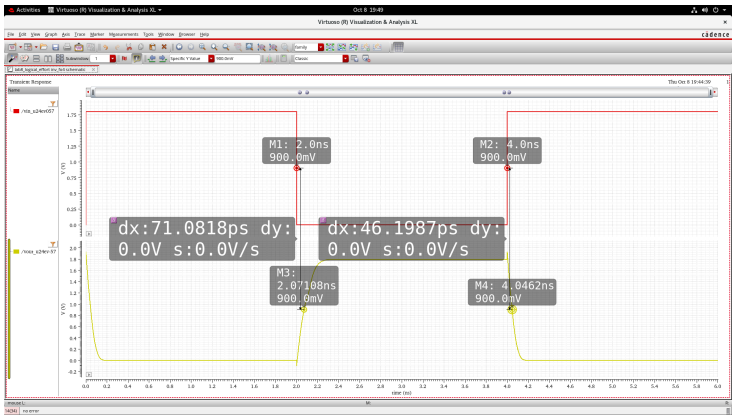


Figure 6.14: CMOS inverter with FO5 transient waveform

$t_{PHL} = 46.1987\text{ps}, \quad t_{PLH} = 71.0818\text{ps}, \quad t_p = 58.6403\text{ps}$

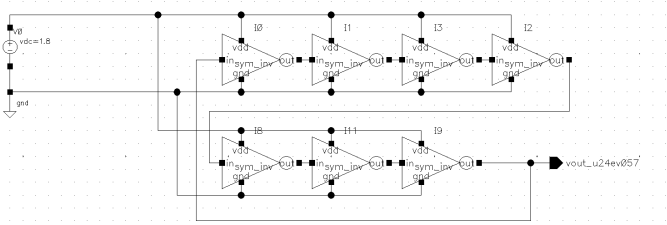


Figure 6.15: 7-stage Ring Oscillator schematic

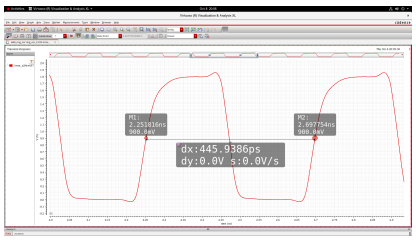


Figure 6.16: 7-stage Ring Oscillator waveform

$T = 445.94\text{ps}, \quad f_{obs} = 2.2425\text{GHz}, \quad N = 7\text{stages}$

7. CONCLUSION

1. Inverter propagation delay scaled linearly with fan-out from FO0 (14.62ps) to FO5 (58.64ps), successfully validating the logical effort delay model.

2. A 7-stage ring oscillator designed for 2.22GHz produced an oscillation frequency of 2.2425GHz ($T = 445.94\text{ps}$), closely matching the objective with $+1.01\%$ error.